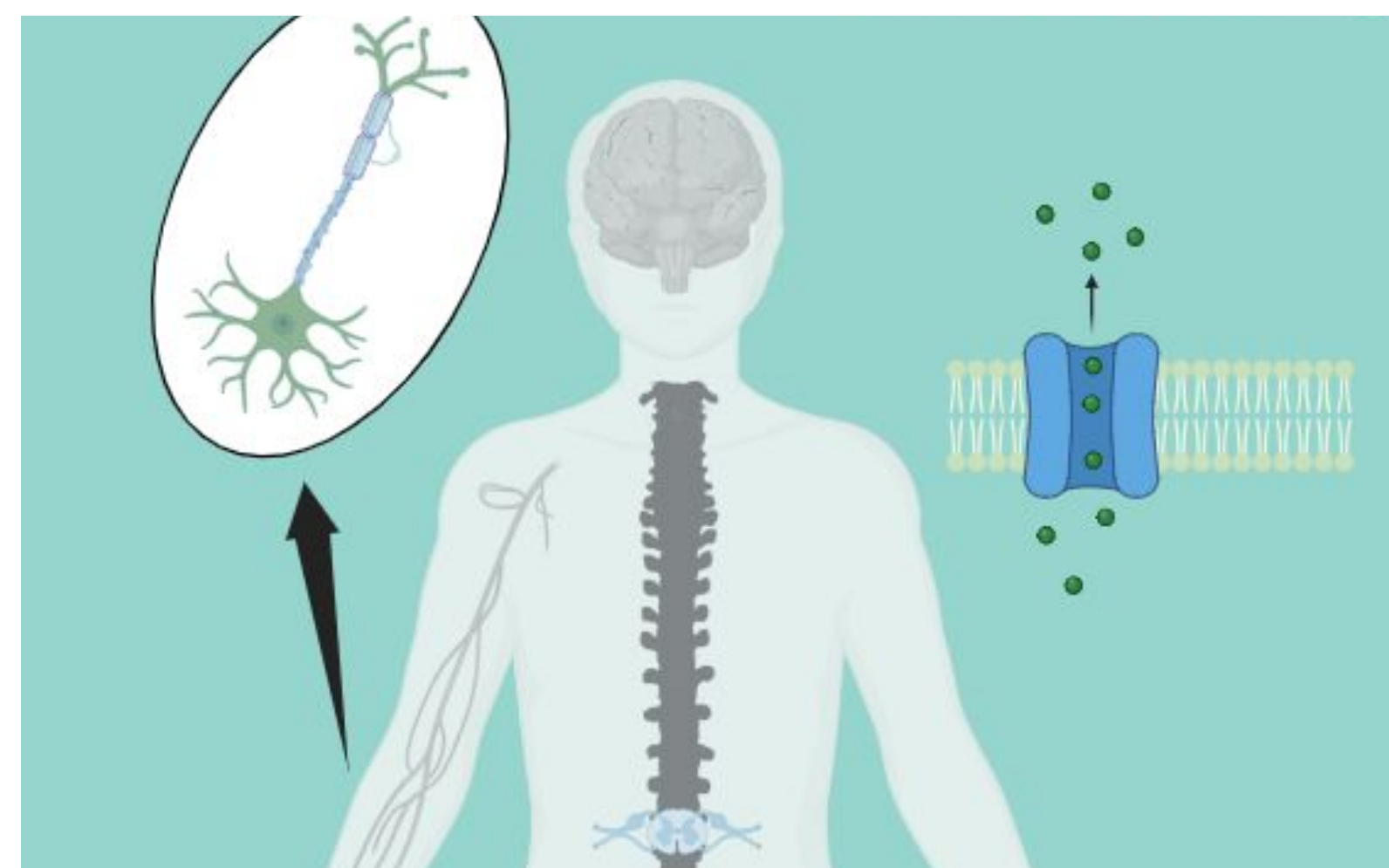


# Determining the Efficacy of A-803467 on Reducing the Sodium Quantity in F11 Cells Modeling Peripheral Neuropathy

## Introduction

**Peripheral neuropathy** is an untreatable chronic condition that can cause pain, affecting an estimated **168 million people** (Morena et al., 2019). It is caused when the myelin sheath around neurons deteriorates and cannot heal, making cells sensitized to chronic pain. High **sodium influx** in nerve cells is associated with high levels of pain.



**Figure 1:** Overview of nervous system made in BioRender by researcher. Demyelination, which leads to hypersensitivity, is deterioration of the protective sheath on the axon of a neuron. Demyelination can be caused by genetics, injuries, or high levels of pro-inflammatory cytokines like TNF- $\alpha$ , leading to peripheral neuropathy.

### Dorsal Root Ganglion (DRG)

Target for understanding development of pain and hyperexcitability of neurons, modeled by F11 cells

### Peripheral Neuron

Demyelination can be seen in the figure in a peripheral neuron (nerve cell not in the central nervous system), and it causes neuropathic pain or numbness; peripheral neurons report to the DRG

### NaV

Voltage-gated sodium channel that opens with the neuron fires an action potential, signaling pain, can maybe be blocked to reduce pain

### A-803467

A-803467 is a NaV channel blocker that has been shown to reduce acute pain in rat models.

**Purpose:** to determine if it can do the same for chronic pain.

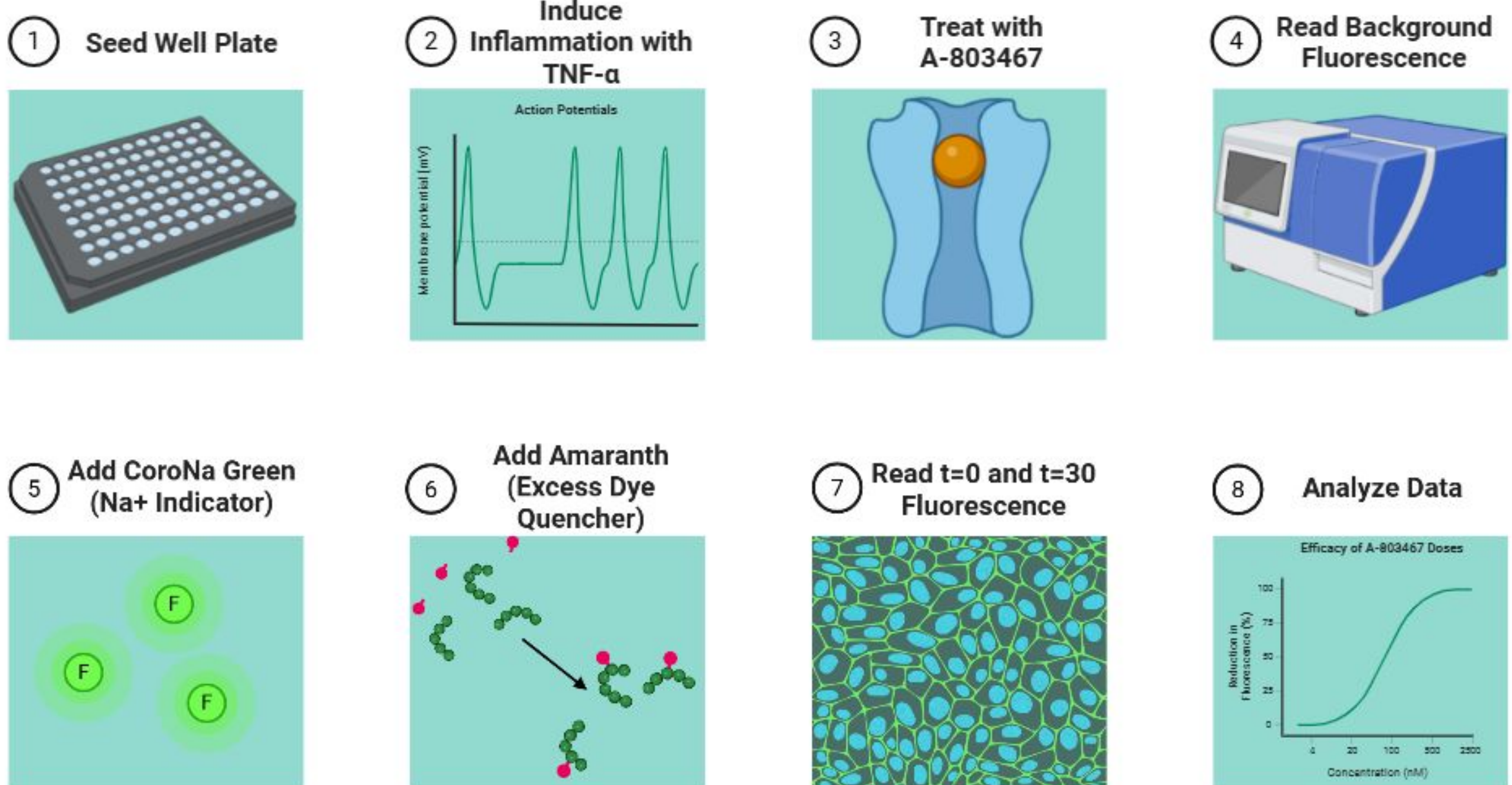
**Hypothesis:** Increasing the concentration of A-803467 administered to cells will decrease the sodium concentration within the cells.

**IV:** Dose of A-803467 (4, 20, 100, 500, 2500 nM).  
**DV:** Change in sodium concentration within the cells, measured with the fluorescent sodium indicator CoroNa Green.

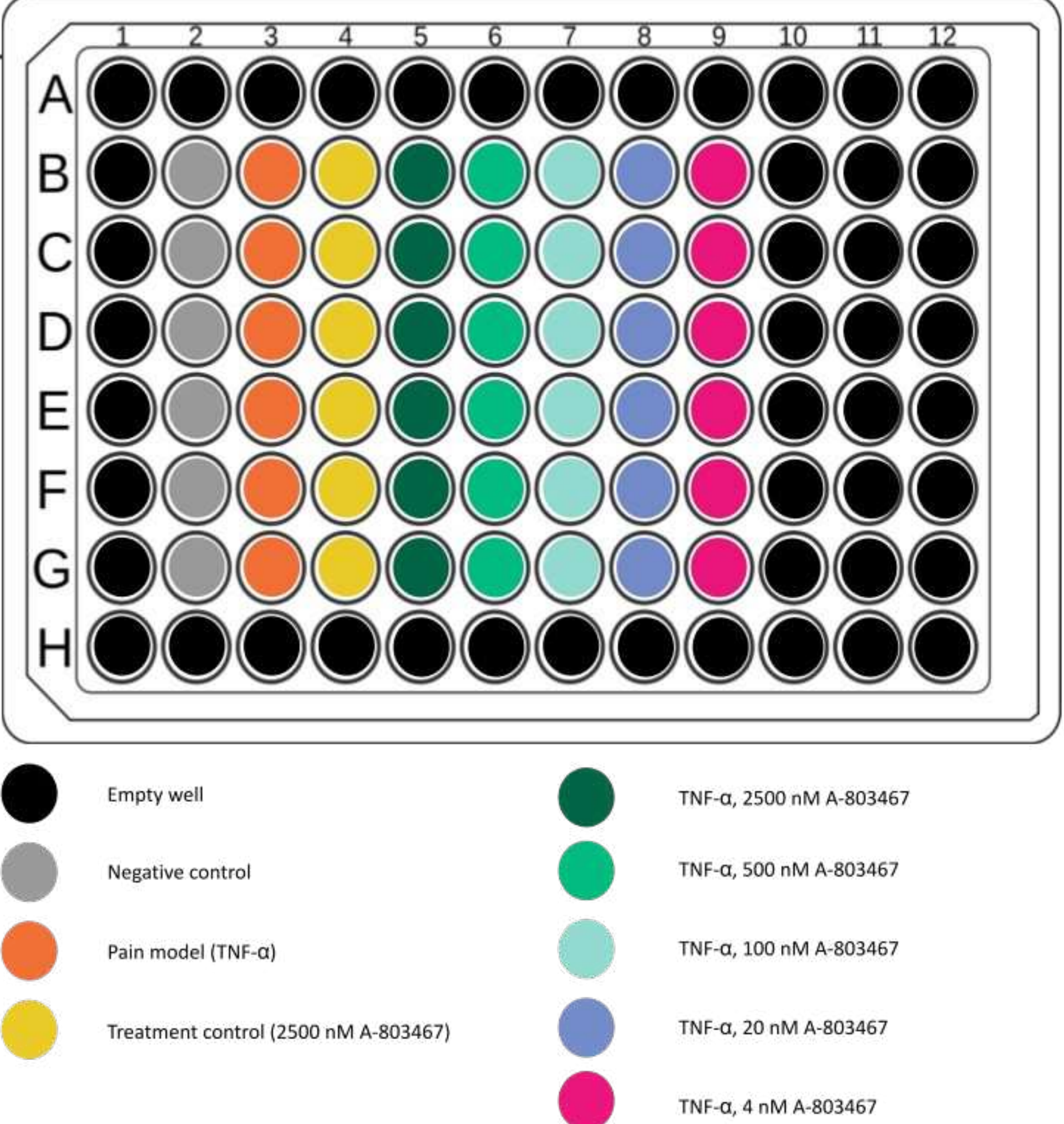
**Controls:** Negative control (no TNF- $\alpha$  or A-803467), chronic pain model (+ TNF- $\alpha$ ), treatment control (- TNF- $\alpha$ )

## Methodology

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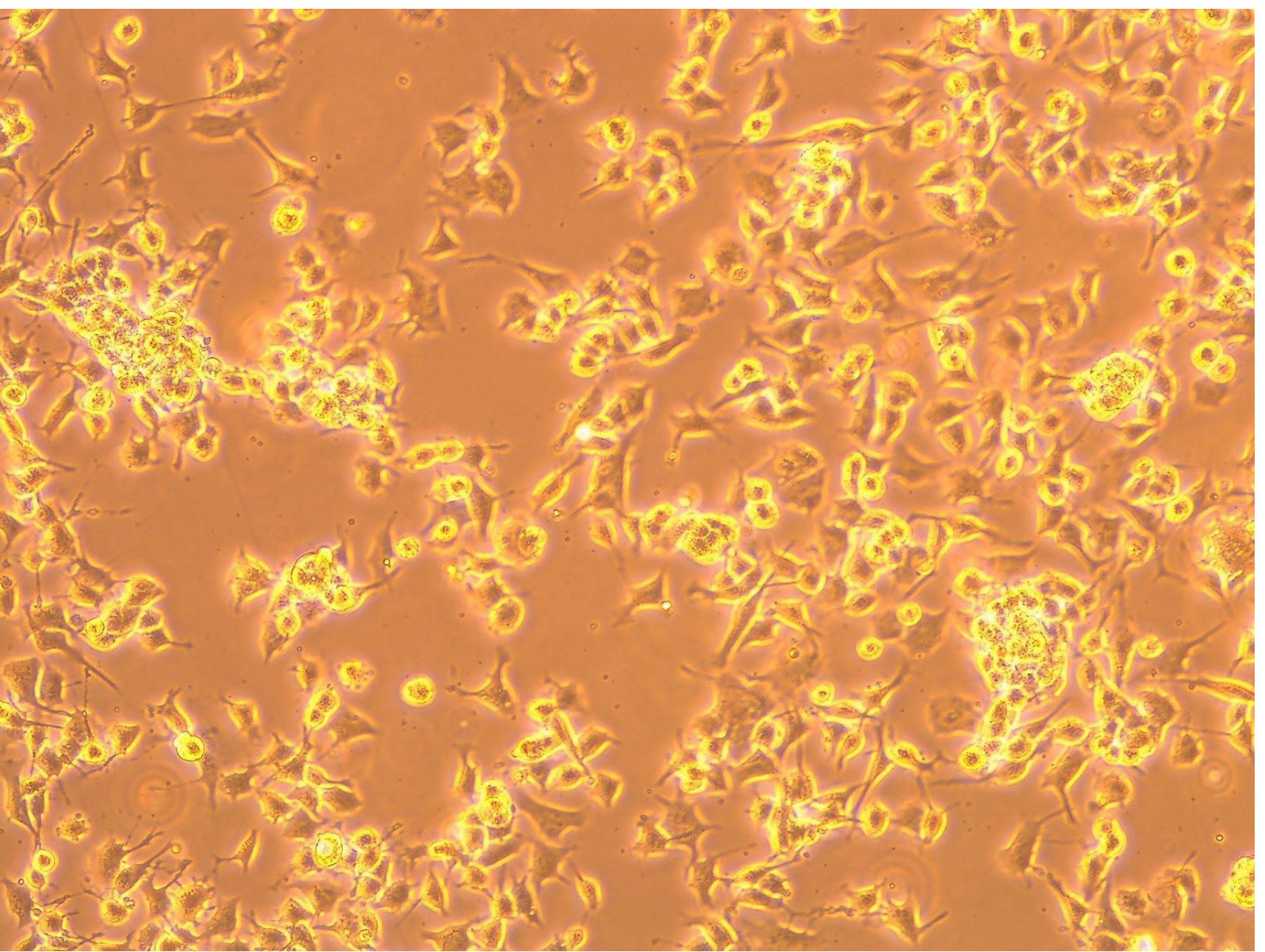


**Figure 2:** CoroNa Green assay to measure the change of intracellular sodium concentration. Cells were seeded and incubated at 37 °C, 5% CO<sub>2</sub> for 48 hours. TNF- $\alpha$  was added for 20 minutes to induce inflammation, and increasing concentrations of A-803467 was added for 10 minutes. CoroNa Green and amaranth were added and fluorescence was measured with a BioTek plate reader at 485 nm ex, 530 nm em (n=12 replicates per group). Image created by researcher using BioRender.

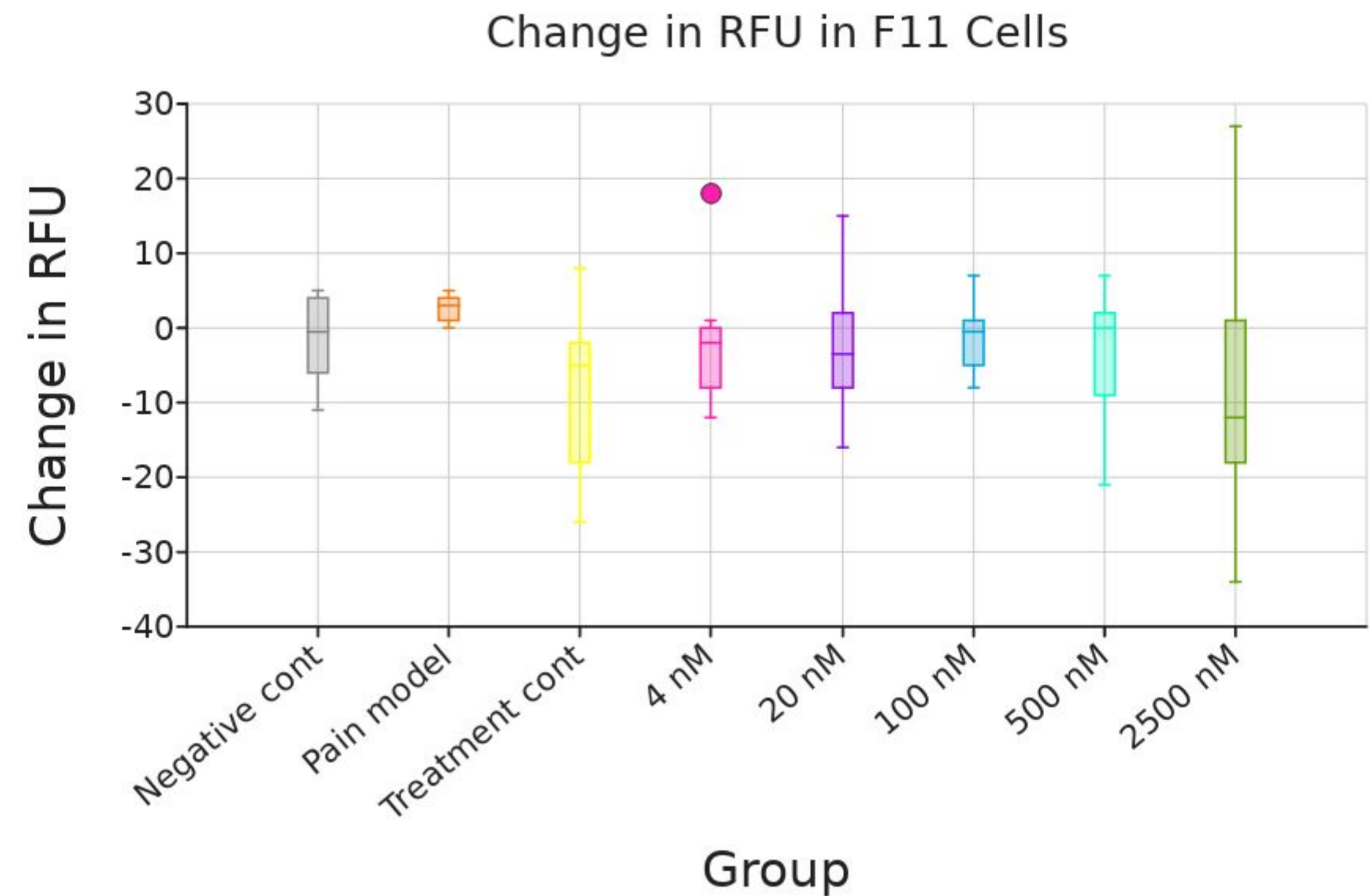


**Figure 3:** Experimental setup of the black 96 well plate. Image created by researcher using Google Drawings.

## Results



**Figure 4:** F11 cells under microscope, 200x magnification. Image taken by researcher using Leica software.



**Figure 5:** Intracellular sodium concentration in cells treated with TNF- $\alpha$  and increasing concentrations of A-803467. A reduction in fluorescence indicates reduced pain. Image created by researcher using DataClassroom.

## Conclusions

The purpose of this project is to determine if A-803467 can reduce the intracellular sodium concentration in a model of neuropathic pain induced by TNF- $\alpha$ . TNF- $\alpha$  has been shown previously to increase the hyperactivity of voltage gated sodium channels which correlates with increased pain. A reduction in intracellular sodium concentration correlates with less hyperactivity of the voltage gated sodium channels in the neurons leading to a reduction in pain.

- After running a Kruskal-Wallis test, it was determined that there was **no statistical significance between the different treatment groups and the controls**.
- A comparison between the negative control and the TNF- $\alpha$  treated pain model did not show an increase in sodium concentration as expected suggesting that the model is not working.
- A comparison between cells treated with TNF- $\alpha$  and TNF- $\alpha$  + 2500 nM A-803467 gave a p value of 0.09. These results suggest that it may be possible to see a difference; however, more data needs to be collected.
- In this model, A-803467 does not have a statistically significant ability to lower the amount of sodium flux in F11 cells.

**Future work:** Optimize the model to show an increase in intracellular sodium after exposure to TNF- $\alpha$ .

**Significance:** More work is needed to understand the potential of A-803467 in treating chronic pain and peripheral neuropathies. Common painkillers and physical therapy are still prescribed to manage the symptoms of neuropathy and other disorders in the nervous system. A-803467 can help increase the rate of nerve healing in organisms with a functioning immune system (Yongchen et al., 2023) but might not decrease the expression of pain.

## References

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